

CSE 604

# Artificial Intelligence

## Chapter 5: Adversarial Search

Adapted from slides available in Russell & Norvig's textbook webpage

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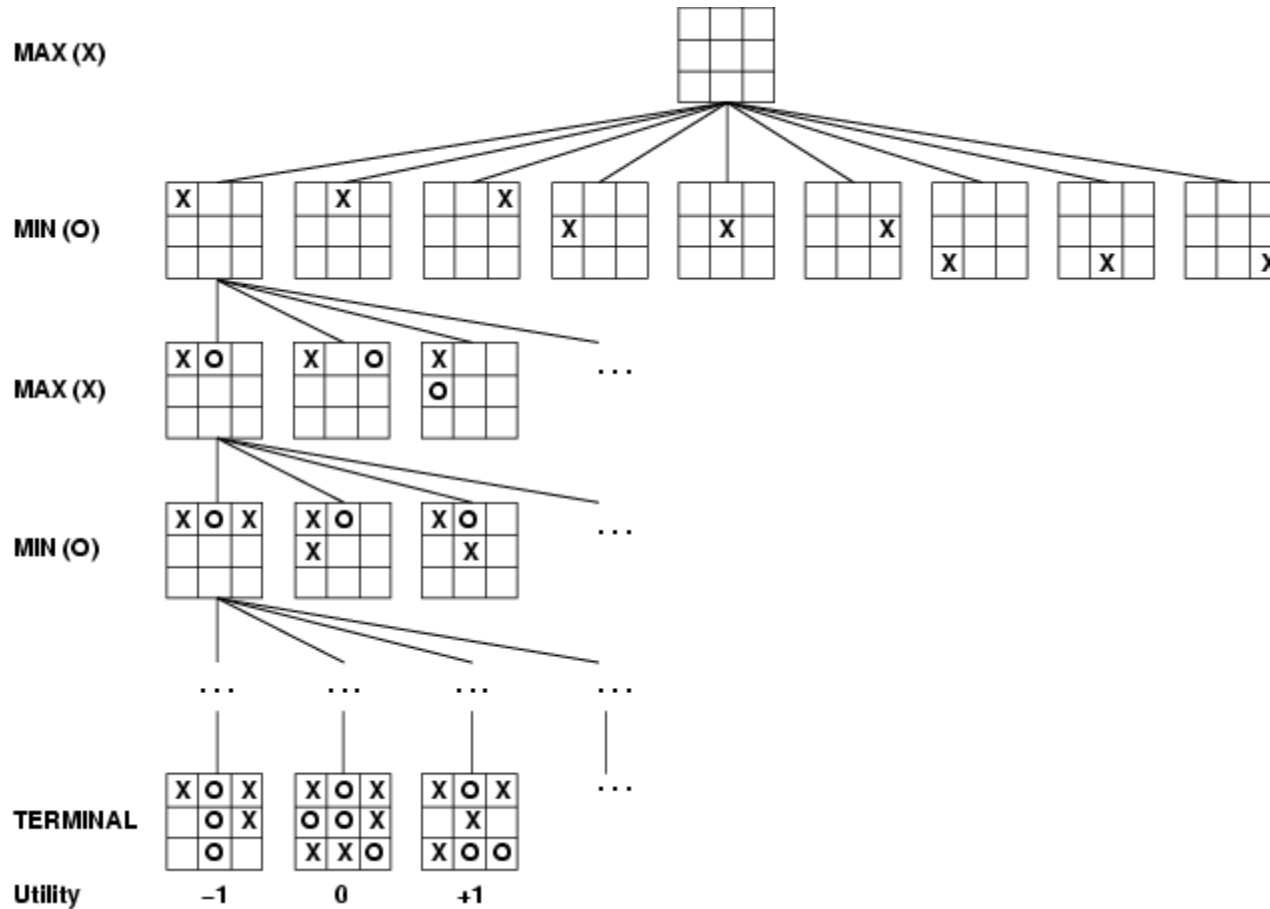
# Outline

- Optimal decisions
- $\alpha$ - $\beta$  pruning
- Imperfect, real-time decisions

# Games vs. search problems

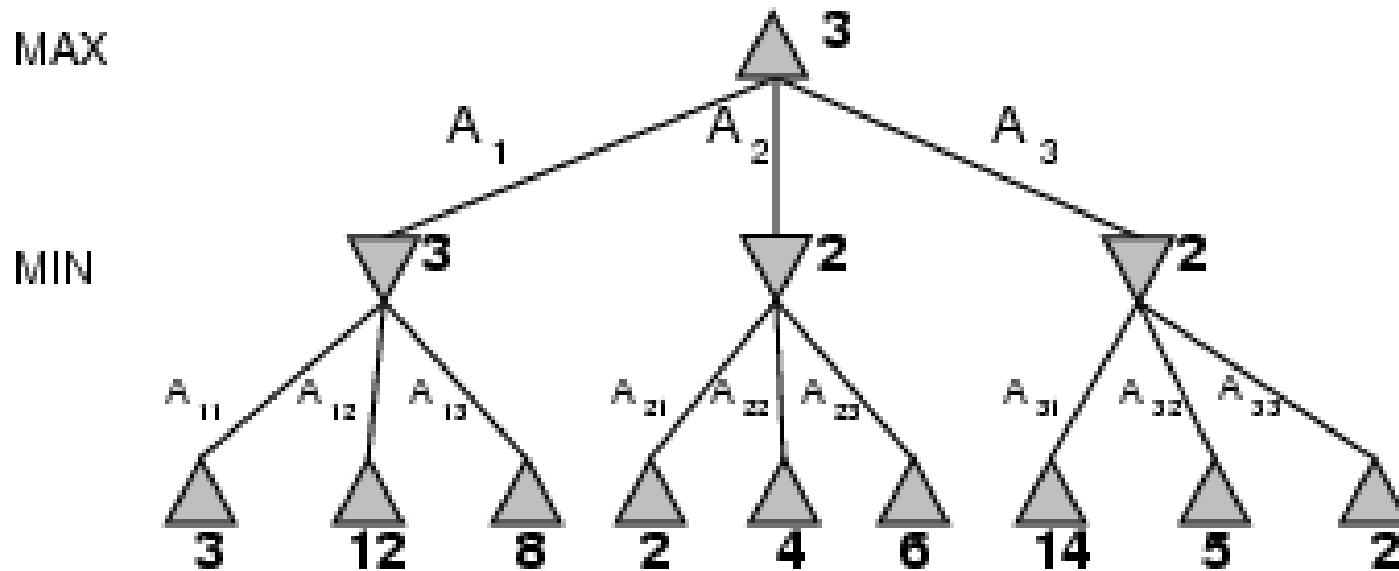
- "Unpredictable" opponent → specifying a move for every possible opponent reply
- Time limits → unlikely to find goal, must approximate

# Game tree (2-player, deterministic, turns)



# Minimax

- Perfect play for deterministic games
- Idea: choose move to position with highest **minimax value**  
= best achievable payoff against best play
- E.g., 2-ply game:



# Minimax algorithm

**function** MINIMAX-DECISION(*state*) *returns an action*

$v \leftarrow \text{MAX-VALUE}(\textit{state})$

**return** the *action* in SUCCESSORS(*state*) with value *v*

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**function** MAX-VALUE(*state*) *returns a utility value*

**if** TERMINAL-TEST(*state*) **then return** UTILITY(*state*)

$v \leftarrow -\infty$

**for** *a, s* in SUCCESSORS(*state*) **do**

$v \leftarrow \text{MAX}(v, \text{MIN-VALUE}(s))$

**return** *v*

---

**function** MIN-VALUE(*state*) *returns a utility value*

**if** TERMINAL-TEST(*state*) **then return** UTILITY(*state*)

$v \leftarrow \infty$

**for** *a, s* in SUCCESSORS(*state*) **do**

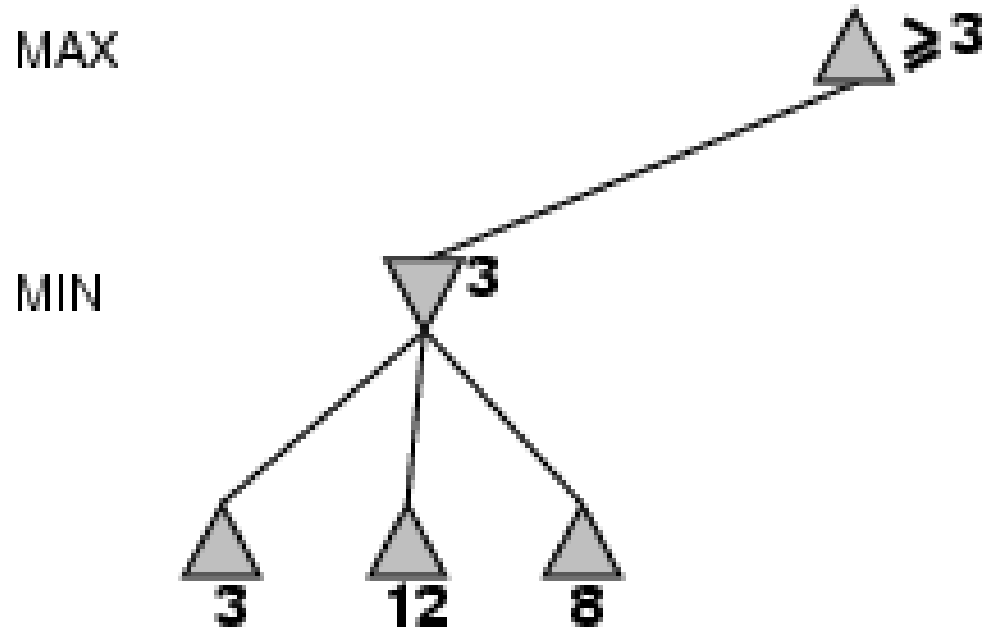
$v \leftarrow \text{MIN}(v, \text{MAX-VALUE}(s))$

**return** *v*

# Properties of minimax

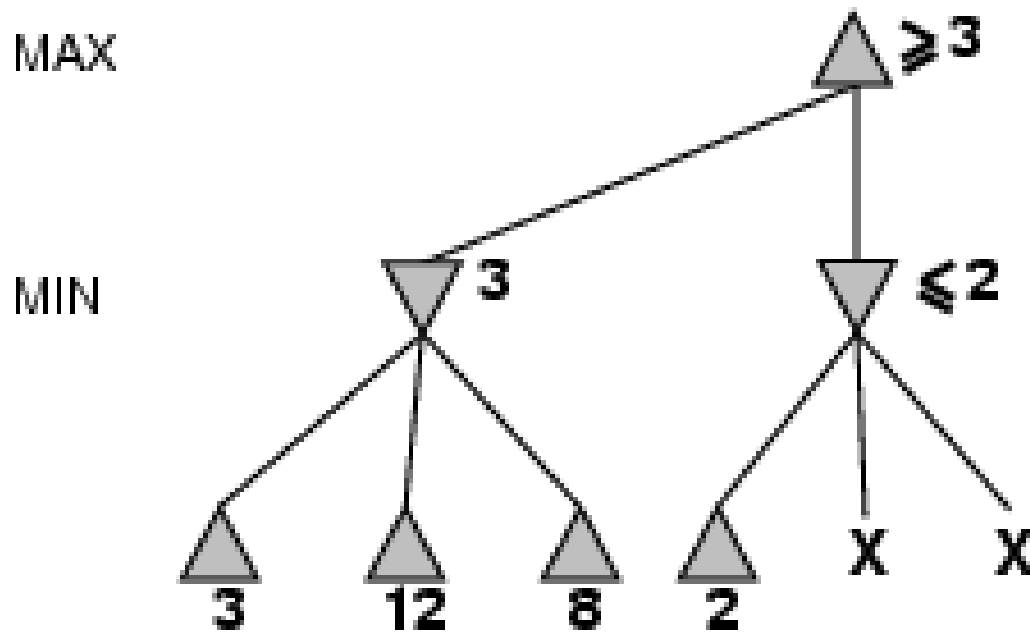
- Complete? Yes (if tree is finite)
- Optimal? Yes (against an optimal opponent)
- Time complexity?  $O(b^m)$
- Space complexity?  $O(bm)$  (depth-first exploration)
- For chess,  $b \approx 35$ ,  $m \approx 100$  for "reasonable" games  
→ exact solution completely infeasible

# $\alpha$ - $\beta$ pruning example

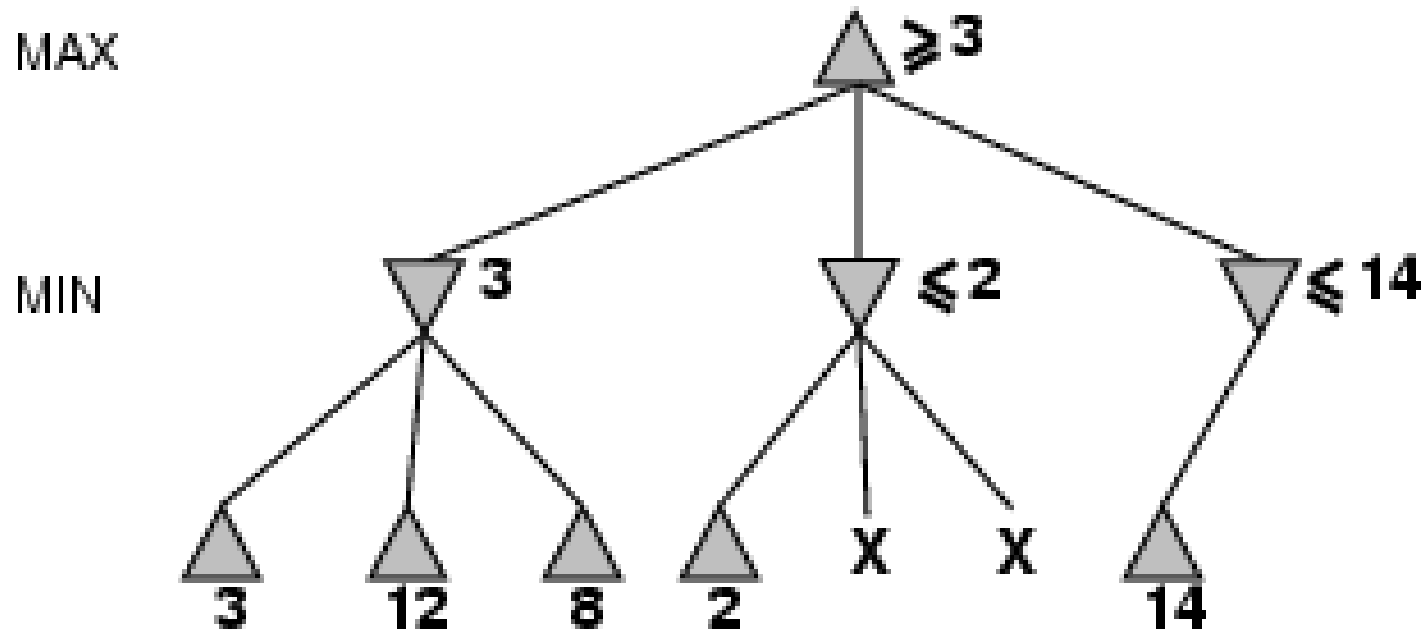




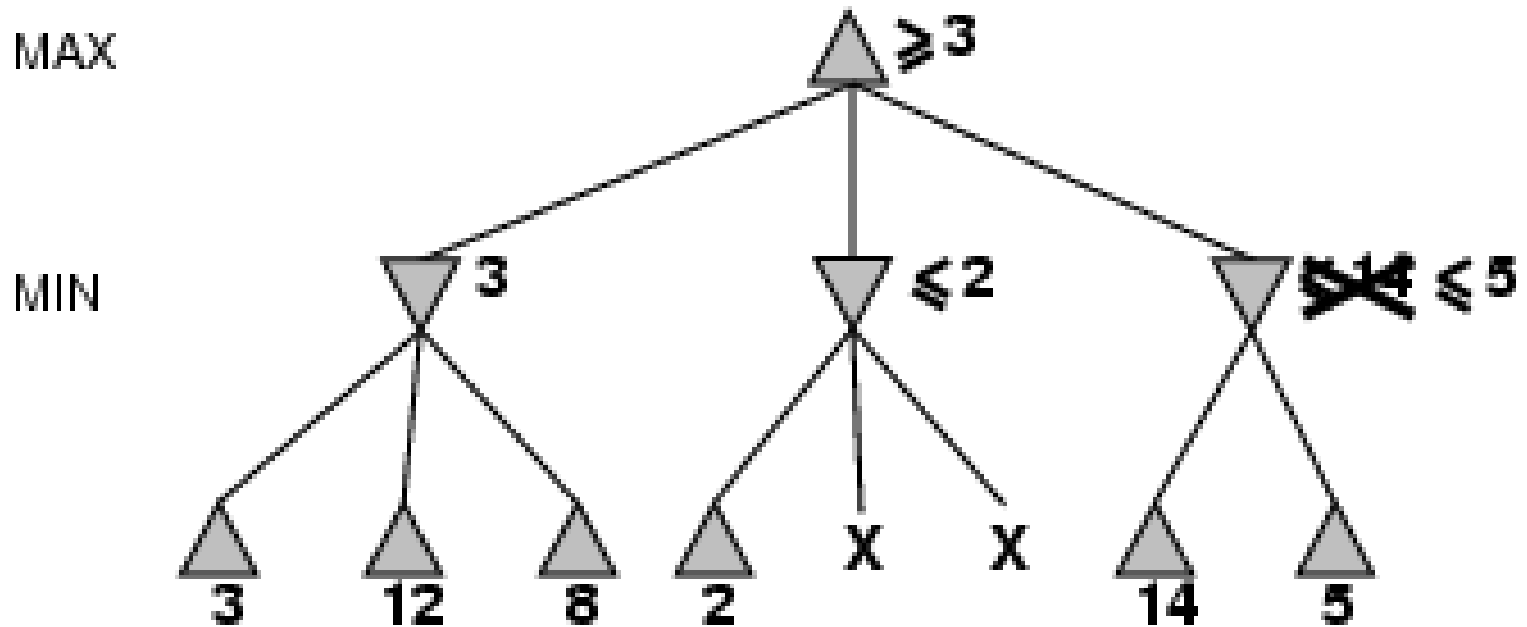
# $\alpha$ - $\beta$ pruning example



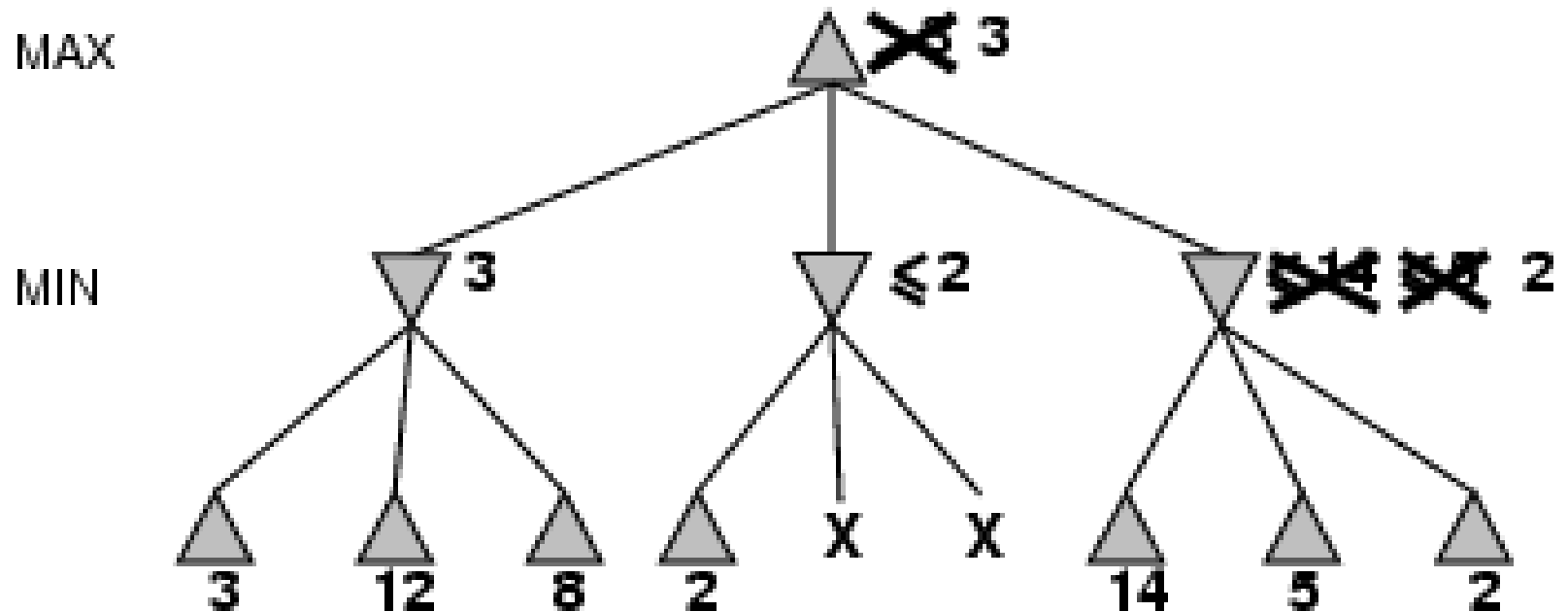
# $\alpha$ - $\beta$ pruning example



# $\alpha$ - $\beta$ pruning example



# $\alpha$ - $\beta$ pruning example

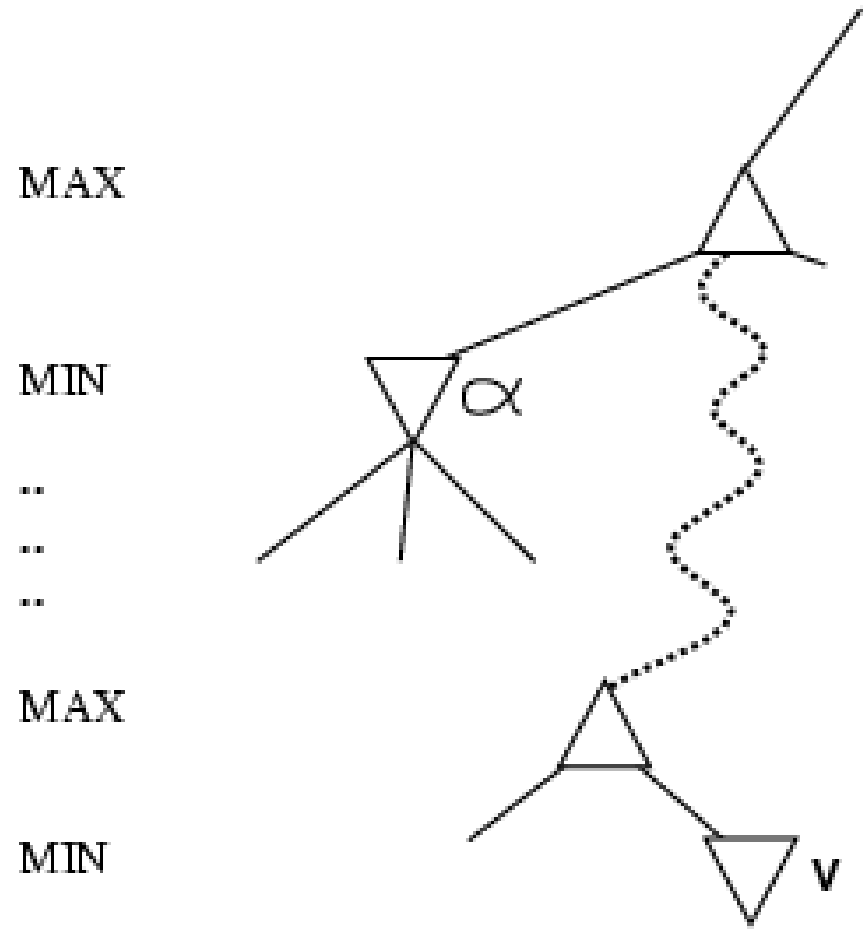


# Properties of $\alpha$ - $\beta$

- Pruning **does not** affect final result
- Good **move ordering** improves effectiveness of pruning
  - Try the moves that are “likely to be best” first
  - E.g., in chess, try captures, threats, forward moves, backward moves in that order
- With "perfect ordering," time complexity  $\approx O(b^{m/2})$ 
  - **doubles** depth of search
- A simple example of the value of reasoning about which computations are relevant (a form of **metareasoning**)

# Why is it called $\alpha$ - $\beta$ ?

- $\alpha$  is the value of the best (i.e., highest-value) choice found so far at any choice point along the path for *max*
- If  $v$  is worse than  $\alpha$ , *max* will avoid it  
→ prune that branch
- Define  $\beta$  similarly for *min*



# The $\alpha$ - $\beta$ algorithm

**function** ALPHA-BETA-SEARCH(*state*) *returns an action*

**inputs:** *state*, current state in game

$v \leftarrow \text{MAX-VALUE}(\text{state}, -\infty, +\infty)$

**return** the *action* in SUCCESSORS(*state*) with value  $v$

---

**function** MAX-VALUE(*state*,  $\alpha$ ,  $\beta$ ) *returns a utility value*

**inputs:** *state*, current state in game

$\alpha$ , the value of the best alternative for MAX along the path to *state*

$\beta$ , the value of the best alternative for MIN along the path to *state*

**if** TERMINAL-TEST(*state*) **then return** UTILITY(*state*)

$v \leftarrow -\infty$

**for**  $a, s$  in SUCCESSORS(*state*) **do**

$v \leftarrow \text{MAX}(v, \text{MIN-VALUE}(s, \alpha, \beta))$

**if**  $v \geq \beta$  **then return**  $v$

$\alpha \leftarrow \text{MAX}(\alpha, v)$

**return**  $v$

# The $\alpha$ - $\beta$ algorithm

```
function MIN-VALUE(state,  $\alpha$ ,  $\beta$ ) returns a utility value
  inputs: state, current state in game
            $\alpha$ , the value of the best alternative for MAX along the path to state
            $\beta$ , the value of the best alternative for MIN along the path to state

  if TERMINAL-TEST(state) then return UTILITY(state)
   $v \leftarrow +\infty$ 
  for  $a, s$  in SUCCESSORS(state) do
     $v \leftarrow \text{MIN}(v, \text{MAX-VALUE}(s, \alpha, \beta))$ 
    if  $v \leq \alpha$  then return  $v$ 
     $\beta \leftarrow \text{MIN}(\beta, v)$ 
  return  $v$ 
```



# Imperfect Real Time Decisions

Even with alpha-beta pruning, it is infeasible to grow the whole game tree!

Standard approach:

- **evaluation function**  
= estimated desirability of position
- **cut off search**  
e.g., depth limit or iterative deepening
- **forward pruning**  
e.g., Beam search

# Evaluation functions

- For chess, typically linear weighted sum of **features**

$$Eval(s) = w_1 f_1(s) + w_2 f_2(s) + \dots + w_n f_n(s)$$

- e.g.,  $w_1 = 9$  with

$f_1(s) = (\text{number of white queens}) - (\text{number of black queens}),$

$w_2 = 5$  with

$f_2(s) = (\text{number of white rooks}) - (\text{number of black rooks}),$

etc.

# Cutting off search

- We can use a modified algorithm *MinimaxCutoff*
- *MinimaxCutoff* is identical to *MinimaxValue* except
  - *Terminal?* is replaced by *Cutoff?*
  - *Utility* is replaced by *Eval*
- Does it work in practice?
  - Suppose we have 100 secs, explore  $10^4$  nodes/sec  
→  $10^6$  nodes per move  
 $b^m = 10^6, b=35 \rightarrow m=4$
- 4-ply lookahead is a hopeless chess player!
  - 4-ply  $\approx$  human novice
  - 8-ply  $\approx$  typical PC, human master
  - 12-ply  $\approx$  Deep Blue, Kasparov

# Deterministic games in practice

- **Checkers:** Chinook ended 40-year-reign of human world champion Marion Tinsley in 1994.
- **Chess:** Deep Blue defeated human world champion Garry Kasparov in a six-game match in 1997. Deep Blue searches 200 million positions per second, uses very sophisticated evaluation, and undisclosed methods for extending some lines of search up to 40 ply.
- **Othello:** human champions refuse to compete against computers, who are too good.
- **Go:** Until recently, human champions refused to compete against computers, who were too bad (in Go,  $b > 300$ ).
  - But in 2016, Google's AlphaGo defeated human world champion Lee Sedol.
  - In 2017, AlphaGo Zero defeated the previous version of AlphaGo 100-0